

SUSHMITHA HALLI SUDHAKARA

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PROJECTS

NYC Taxi Demand Forecaster - Serverless ML Pipeline | [GitHub](#)

Jan 2025 - Apr 2025

Hourly taxi-demand forecasting with a feature store and automated retraining

- **Reduced test MAE by 26%, from a 3.37 best-baseline to 2.49**, by benchmarking naive baselines against XGBoost (2.60) and LightGBM (2.54), then tuning LightGBM hyperparameters with Optuna over a TimeSeriesSplit cross-validation objective to avoid look-ahead leakage.
- **Automated hourly feature ingestion and inference end to end** by implementing two chained GitHub Actions workflows - a cron-scheduled feature pipeline writing to a Hopsworks online feature store, triggering an inference pipeline that loads the model from the registry and writes predictions back as a feature group.
- **Deployed a live Streamlit forecasting and monitoring app** backed by the feature store, joining stored predictions against observed ride counts to track model error over time, with the training and inference code packaged as a Poetry-managed src/module using custom scikit-learn transformers for lag and temporal features.

AWS Serverless MLOps Pipeline - Vehicle Image Classification | [GitHub](#)

Sep 2024 - Oct 2024

Vehicle-type classification for delivery routing

- **Image Classification & Data Preparation:** Trained a bicycle-vs.-motorcycle image classifier with Amazon SageMaker's built-in image-classification algorithm (CNN); decoded 3,073-integer CIFAR-100 records into 32×32×3 RGB arrays, staged PNG images and TSV metadata in Amazon S3, and trained for 30 epochs on an ml.p3.2xlarge instance, achieving 81% validation accuracy.
- **Serverless ML Inference Pipeline:** Deployed the model to a SageMaker endpoint and orchestrated inference with AWS Step Functions and three Lambda functions for data generation, image classification, and confidence filtering; routed predictions below a **75% confidence threshold** for review.
- **Model Monitoring:** Configured SageMaker **DataCaptureConfig** to capture **100% of inference requests and responses** to S3 and built custom visualizations over the captured data to track model behavior and prediction quality.

Customer Churn Prediction | [GitHub](#)

Sep 2024 - Oct 2024

Churn risk prediction with LLM-generated explanations and retention actions

- **Improved minority-class recall by 10%** by addressing class imbalance with **SMOTE** and evaluating ensemble-based churn prediction models.
- Built a **Streamlit application for real-time churn scoring**, combining XGBoost, Random Forest, and KNN predictions to estimate customer churn probability and display model-level outputs.
- Integrated Llama models through the Groq API to generate natural-language explanations grounded in customer attributes and model feature importance, followed by personalized retention emails based on predicted churn risk.

Reproducibility Study: "Generating Long Sequences with Sparse Transformers" | [GitHub](#)

Apr 2024 - May 2024

Graduate deep learning course project (CS 895, ODU); Owned the image-modality implementation

- **Reproduced the paper's CIFAR-10 result to within 0.13 bits/dimension - 2.93 against the reported 2.80**, on roughly 1/8 the training budget (15 epochs vs. 120), by reimplementing the fixed sparse attention factorization from Child et al. (2019) for image data and training a 128-layer, 2-head, 256-embedding model on 4 NVIDIA A100s.
- **Isolated training duration and model capacity, rather than the sparse attention mechanism itself, as the source of the replication gap**, by holding learning rate, dropout, layers, heads, and embedding size fixed to the paper's reported configuration and varying only what the available compute allowed - the original used 64 V100s over 7 days.

Bike Sharing Demand Prediction with AutoGluon | [GitHub](#)

Apr 2024 - May 2024

Demand forecasting for bike-sharing operations

- **Improved Kaggle score from 1.80462 to 0.49549** through iterative temporal feature engineering and model evaluation using AutoGluon **TabularPredictor**.
- **Achieved a 0.49549 score within a 10-minute training window** using **best_quality** preset, automated hyperparameter optimization, and multi-model stacking/weighting with RMSE evaluation.
- **Analyzed feature impact across iterations**, finding temporal features contributed **92% of total improvement**, while one-hot encoding reduced performance by **1.7%**.

WORK EXPERIENCE

Technical Sales Enablement Specialist (AI & Automation Projects), Decisions LLC

Jul 2024 - Present

- **LLM Prompt Engineering (GTM Automation):** Designed and iterated production prompts that extract sales opportunities and classify use cases from call transcripts; enforced strict JSON-only structured outputs, added confidence scores and classification rationale, and debugged over-restrictive filtering logic to reduce false negatives. Validated prompt behavior across GPT-4, GPT-5, and open-source models, and version-tracked prompt iterations in GitHub.

- **LLM Evaluation & Failure Analysis:** Investigated inconsistent GPT-5 ChatCompletion behavior, including empty/truncated responses and tool-flow failures; reviewed reasoning, tool-calling, and output-token behavior, then added structured success/fallback JSON responses for empty, malformed, incomplete, and failed tool-flow results. Tested the updated prompt in the internal QA environment.
- **RAG & AI Agent Workflow:** Developed and tested prompts and tool flows for an internal AI assistant over Lunch & Learn data; refined tool selection, date-based query handling, and Chat Completion message history, and debugged vector retrieval by tracing embeddings through the PostgreSQL/pgvector pipeline.
- **Rule Extraction Automation:** Created and tested rule sets generated from document extracts; iterated on system prompts to improve extraction accuracy and structured outputs, and flagged missing inputs/variables in specs to stakeholders.
- **Doc360 Automation (Decisions Flow):** Built a flow to pull Doc360 article links and extract article overviews, supporting documentation audits, tracking, and content updates.
- **Hands-on Feature Validation & Documentation:** Authored end-to-end documentation for AI platform features (agents, chat completion, embeddings, MCP servers) and the Deployment Server based on hands-on testing of expected vs. actual behavior; partnered with Product, Engineering, QA, and Support during fast-paced releases.

Machine Learning Researcher, VMASC, Norfolk, VA

Jan 2024 - Jan 2025

- Developed a Native AI framework for cyber-physical energy systems using ensemble learning (Random Forest, XGBoost) and LSTM networks, achieving 99.97% fault-detection recall and reducing DER simulation downtime by 30%; research accepted and presented at IEEE IWCMC 2025.
- Introduced Gini-Permutation feature selection and temporal encoding, improving cross-system generalization by 22%.
- Applied FedAvg and GANs to enhance smart-grid security on imbalanced datasets, achieving 81% accuracy, 85% F1, 78% precision, and 94% recall.
- Co-authored two research papers on federated learning and SHAP-based model interpretability for smart-grid and UAV-swarm security (ICAI'24; IEEE OJ-ITS).

Graduate Research Assistant, Old Dominion University, Norfolk, VA

Sep 2023 - Dec 2023

- **Image Segmentation:** Refined SAM (Segment Anything Model) for menu-card digitization using GIMP-assisted image segmentation and Python automation scripts for structured data extraction.
- Integrated SAM with PyTorch and Hugging Face Transformers for efficient image-processing and model-training workflows.
- Web-scraped 10,000+ Reddit entries with Python and BeautifulSoup to build a robust sentiment-analysis training dataset.

Data Science Intern, NightHack, Bengaluru, India

Jun 2023 - Aug 2023

- **OCR & Text Extraction:** Applied PaddleOCR and Transformers for text extraction and tag classification across 2,000+ titles, scaling the solution to 10,000 titles and significantly improving extraction accuracy.
- Performed exploratory data analysis and managed large-scale datasets in MongoDB for reliable storage and retrieval.

Software Engineer – Cloud Services, LTIMindtree, Bengaluru, India

Sep 2019 - Jul 2022

- Built a Timecard web app integrated with Oracle Fusion Cloud (REST APIs), replacing a slow, error-prone workflow and cutting time-to-submit, rework, and support effort.
- Automated supplier invoice payments with UiPath and implemented SSO for Oracle Fusion Cloud; developed custom HDL templates, fast formulas, and SQL reports for bulk data management and process improvement.

SKILLS

Machine Learning & Computer Vision: Python, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, SAM (Segment Anything Model), PaddleOCR, CNNs, AutoGluon, SHAP, NumPy, Pandas

LLM & Generative AI: Prompt engineering, structured JSON outputs, embeddings, AI agents & tool use, MCP servers, OpenAI GPT-4/GPT-5, Llama 3.2 (GROQ API)

Cloud & DevOps: AWS (S3, Lambda, Step Functions, SageMaker, EC2), Git, GitHub Actions (CI/CD), Streamlit, Oracle Fusion Cloud

Data & Tools: MySQL, PostgreSQL, MongoDB, Matplotlib, Plotly, Tableau, Jupyter, Postman, VS Code, Java, SQL

Product & Delivery: Hands-on feature validation (expected vs. actual behavior), requirements clarification, troubleshooting and edge-case analysis, cross-functional collaboration, clear technical communication for technical and non-technical audiences

PUBLICATIONS

- S. H. Sudhakara, M. S. Munir, M. Rahman and S. Shetty, "Native AI-based Predictive Operational Resiliency in Cyber-Physical Energy Systems," IWCMC 2025, Abu Dhabi, UAE, pp. 1258-1263, doi: 10.1109/IWCMC65282.2025.11059490. | [Link](#)
- S. Halli Sudhakara and L. Haghnegahdar, "Security Enhancement in AAV Swarms: A Case Study Using Federated Learning and SHAP Analysis," in IEEE Open Journal of Intelligent Transportation Systems | [Link](#)
- Sudhakara, S.H., Haghnegahdar, L., GhasemiGol, M., Takabi, D. (2025). "Utilizing Federated Learning and SHAP for Predictive Analysis in Smart Grid Security." CSCE 2024, CCIS vol. 2251, Springer, Cham. | [Link](#)

EDUCATION

M.S. in Computer Science, Old Dominion University - GPA 3.88

Aug 2022 - Dec 2024

B.E. in Information Technology, Visvesvaraya Technological University - GPA 3.4

Jun 2015 - Aug 2019